PROJECT 1 · GIS AND REMOTE SENSING

Agricultural and water drought in the Upper Litani Basin, Lebanon, 2001–2025

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The Upper Litani Basin is the farming heart of the Bekaa valley and feeds Lake Qaraoun, Lebanon's largest reservoir. This project uses 25 years of satellite data to answer four questions: which growing seasons were drought years, what drove them, how much the lake shrank in dry years, and which cadastres are hit most often.

71 %
of cropland in drought in 2008, the worst season of 2001–2025

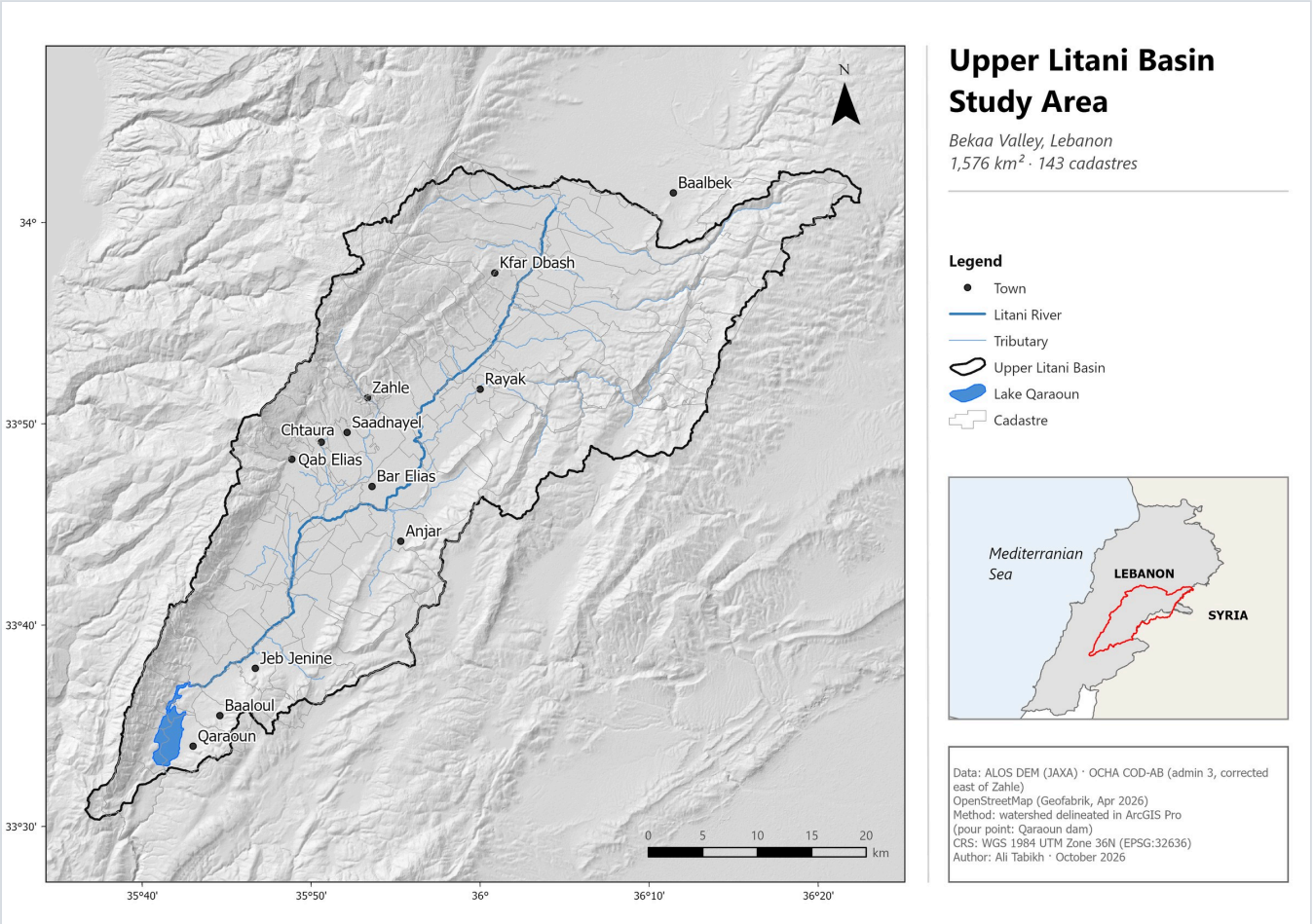
r = −0.96
drought area vs the temperature index: heat, more than rain, sets the yearly drought

−61 %
Lake Qaraoun area, dry autumn 2014 vs wet autumn 2019

8
drought hot spots, all in the rain-fed north of Baalbek caza

Study area

The basin runs 1,576 km² from the Litani source west of Baalbek to the Qaraoun dam, between Mount Lebanon and the Anti-Lebanon. It holds 143 cadastres and about 300 km² of cropland. The north is mostly rain-fed; the south-west plain (West Bekaa, south Zahle) is more irrigated. The outline was delineated from a DEM in ArcGIS Pro with the dam as pour point.



Map 1 · Upper Litani Basin, 143 cadastres. One OCHA admin-3 polygon that wrongly covered about 20 cadastres east of Zahle was replaced by 21 corrected boundaries.

Data and method

Data	Resolution	Used for
MODIS MOD13Q1 NDVI	250 m, 16 days	Vegetation Condition Index (VCI)
MODIS MOD11A2 land surface temperature	1 km, 8 days	Temperature Condition Index (TCI)
CHIRPS daily rainfall (Funk et al., 2015)	~5.5 km	Rainfall per hydrological year (Sept–Aug) and anomaly
Landsat 5, 7, 8, 9 (Collection 2, L2)	30 m	Lake Qaraoun water area (MNDWI > 0; Xu, 2006)
ESA WorldCover 2021	10 m	Cropland mask (> 50 % of each MODIS pixel)
OCHA COD-AB admin 3, DEM	vector, 12.5 m	Cadastres (corrected east of Zahle), basin outline

1. **Google Earth Engine:** VCI, TCI and the Vegetation Health Index (VHI = 0.5 VCI + 0.5 TCI; Kogan, 1995) for the March–May growing season, over cropland only. A pixel is in drought when VHI < 40. Rainfall, lake area and indicators per cadastre were computed in the same scripts.
2. **Python:** charts and trend tests (Mann-Kendall, Sen's slope) on each yearly series.
3. **SPSS:** Pearson and Spearman correlation, simple and multiple linear regression (n = 25 years).
4. **ArcGIS Pro:** cadastre correction and joins with arcpy, drought frequency per pixel, Hot Spot Analysis (Getis-Ord Gi*; Ord & Getis, 1995; FDR-corrected) on the 87 cadastres with at least 5 % cropland, and all maps.

1 Greenness alone can mislead: 2003 was cold, not dry

2003 had the lowest vegetation index of the 25 years (VCI 30), yet it was the second-wettest year (+37 % rain) with a cool spring (TCI 81). With temperature included, VHI is 56 and only 17 % of the cropland was in drought: the late greening came from cold weather, not from a lack of water.

This is why the project uses VHI and not NDVI or VCI alone.

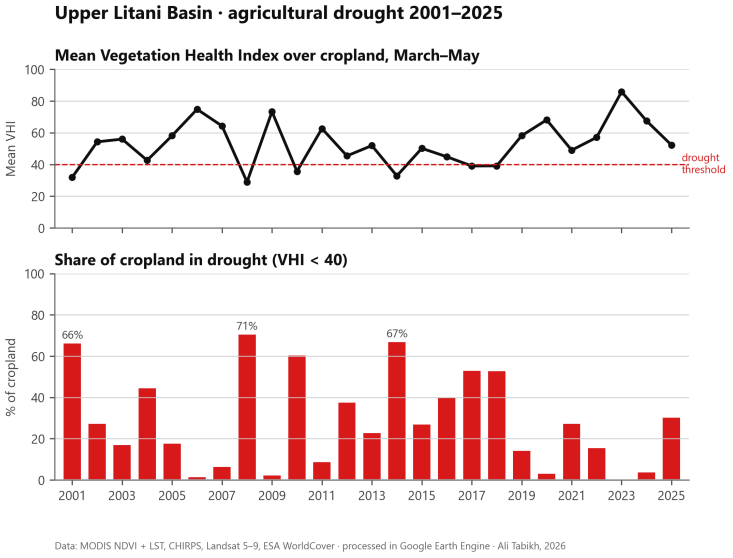


Chart 1 · Mean VHI and share of cropland in drought, 2001–2025.

The six worst seasons

Season	Cropland in drought	Mean VHI	Rain vs average	Spring land temperature	Autumn lake area
2008	71 %	28.9	−32 %	29.9 °C	4.6 km²
2014	67 %	32.8	−20 %	29.4 °C	3.2 km²
2001	66 %	31.9	−17 %	29.7 °C	3.8 km²
2010	61 %	35.7	+7 %	29.0 °C	5.9 km²
2017	53 %	39.1	−25 %	28.6 °C	5.6 km²
2018	53 %	39.1	+8 %	29.0 °C	4.4 km²
Mean 2001–2025	29 %	52.9	591 mm	27.3 °C	5.8 km²

Table 1 · All six seasons had a spring 1.3–2.6 °C warmer than average; two of them (2010, 2018) had normal rain.

2 The worst seasons were 2008, 2014 and 2001, and heat matters as much as rain

Share of cropland in drought: 2008 71 %, 2014 67 %, 2001 66 %, 2010 61 %, 2017 and 2018 53 % (Table 1). Most of these years had a rain deficit (2008 −32 %, 2014 −20 %, 2017 −25 %), but 2010 and 2018 had normal rain and a very hot spring.

Each 100 mm less rain adds about 9 points of cropland in drought (R^2 = 0.21, p = 0.02): rain explains only a fifth of the variation. Drought area follows the temperature index far more closely (r = −0.96) than the greenness index (r = −0.40), and dry years are also hot years (rain vs spring temperature r = −0.45).

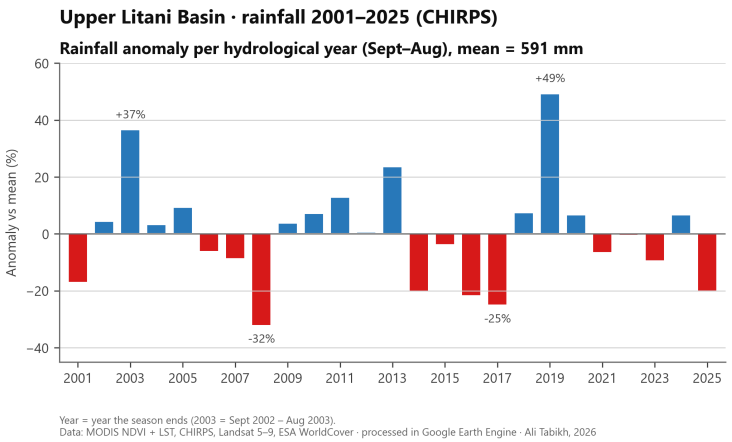


Chart 2 · Rainfall anomaly per hydrological year (CHIRPS).

3 Lake Qaraoun mirrors the drought

The lake covers on average 9.6 km² in spring and 5.8 km² in autumn. Rain explains almost half of the autumn area (R^2 = 0.43): each 100 mm more rain adds about 0.7 km². A small autumn lake goes with a large share of cropland in drought (r = −0.72, p < 0.001).

The autumn lake was smallest in 2014 (3.23 km²) and largest in 2019 (8.35 km²), a 61 % difference (Map 2).

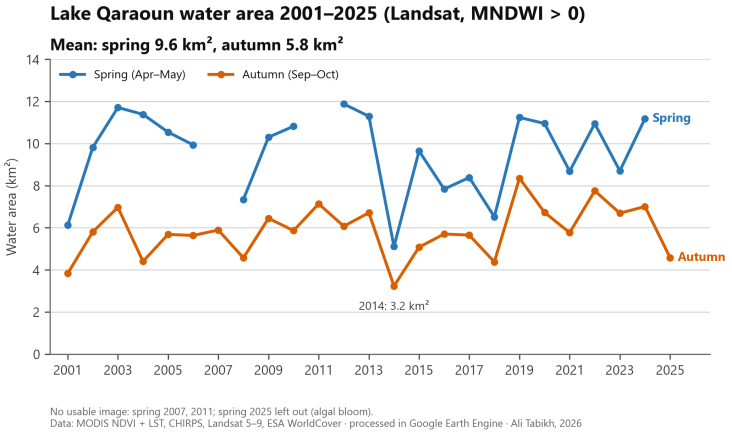
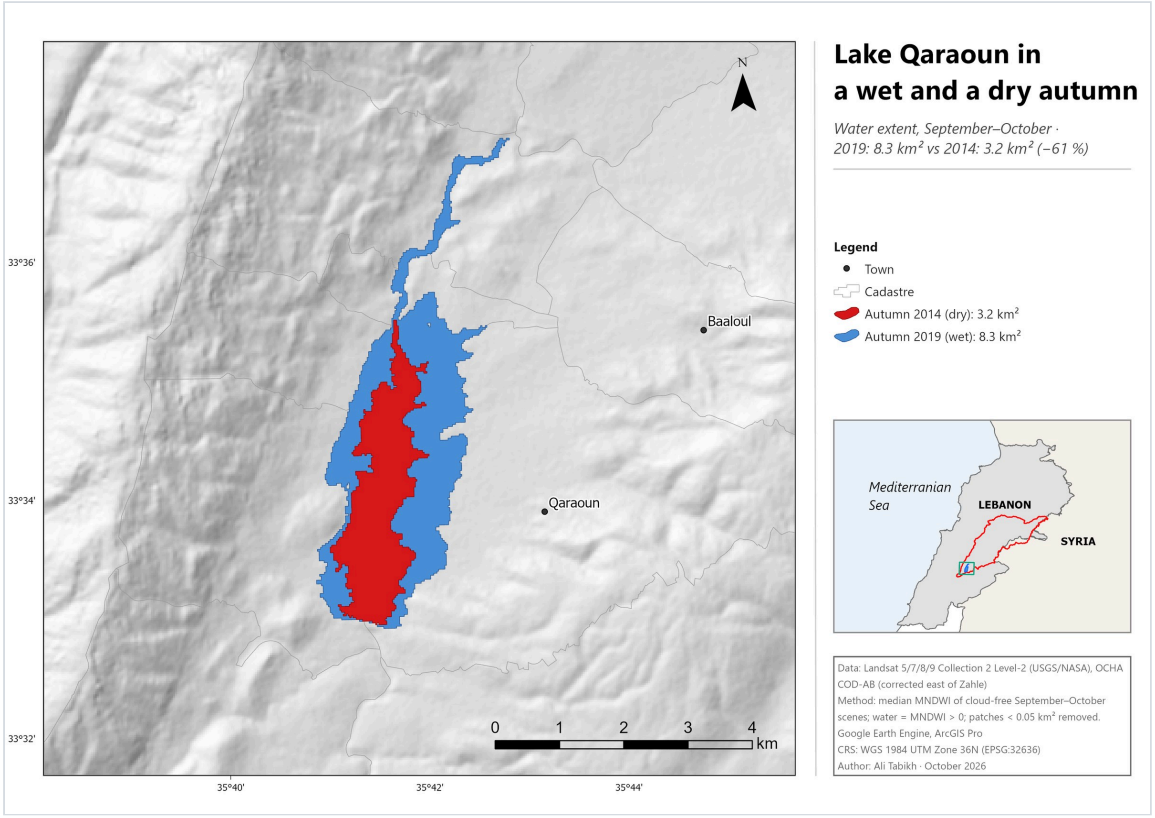


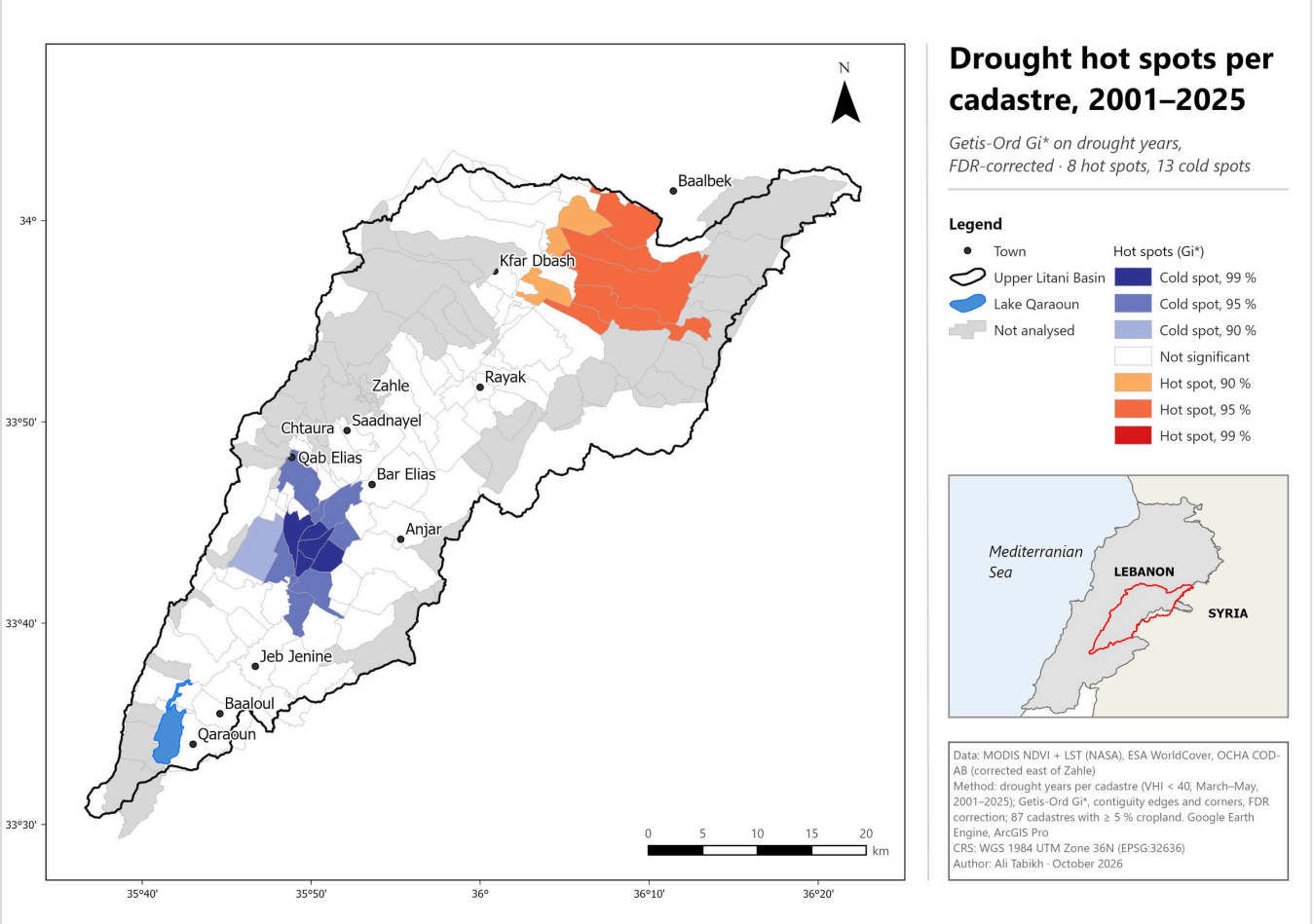
Chart 3 · Lake Qaraoun water area, spring and autumn.



Map 2 · Lake Qaraoun in autumn (September–October) 2019 and 2014, Landsat MNDWI.

4 The rain-fed north is the most exposed

Cropland in Baalbek caza was in drought in 8.6 of the 25 seasons on average, against 7.4 in Zahle caza and 6.3 in West Bekaa. The hot spot analysis confirms the pattern: 8 hot spots, all in north Baalbek caza (Hour Taala, Khodr Baalbek, Brital, Taibet Baalbek and Douris at 95 %; Majdaloun, Hizzine and Haouch Snaid at 90 %), with 10–11 drought years in the most exposed. The 13 cold spots lie in the irrigated plain of West Bekaa and south Zahle (5 at 99 %).



Map 3 · Hot and cold spots of drought years per cadastre (Getis-Ord Gi*, FDR-corrected, 87 cadastres).

5 No clear long-term trend, but 2025 was a dry year

None of the series has a significant trend over 2001–2025 (Mann-Kendall, $p > 0.15$): VHI, drought area, rainfall, temperature and lake area swing from year to year rather than moving in one direction. At pixel level, cropland was in drought in 7.2 of the 25 seasons on average, and 12 % of it in 11 seasons or more. 2025 had 20 % less rain than average, yet only 30 % of the cropland was in drought, far less than in 2008 or 2014; irrigation may have buffered the crops, which this project does not test.

Limits

- MODIS pixels are 250 m (temperature 1 km) and CHIRPS ~5.5 km: small plots are smoothed. One cropland mask (WorldCover 2021) is used for all 25 years.
- VHI contains land surface temperature, so its link with temperature is partly built in.
- Lake area: no usable spring image in 2007 and 2011; spring 2025 left out because an algal bloom hid the water from MNDWI.

How this compares with other studies

Faour et al. (2015) mapped VHI across Lebanon with Landsat for 1982–2014 and also linked vegetation drought mainly to rising temperature, which matches the strong heat signal found here. Awad (2026) tracked Lake Qaraoun with Landsat 7–9 and MNDWI in Google Earth Engine and reports a fall in mean yearly volume from 2003 to 2025; this project finds no significant trend in autumn lake area over 2001–2025. The two results measure different things (yearly volume vs autumn surface area) over slightly different periods.

References

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Tools: Google Earth Engine · ArcGIS Pro and arcpy · Python (pandas, matplotlib, pymannkendall) · SPSS. Scripts in the project repository.